

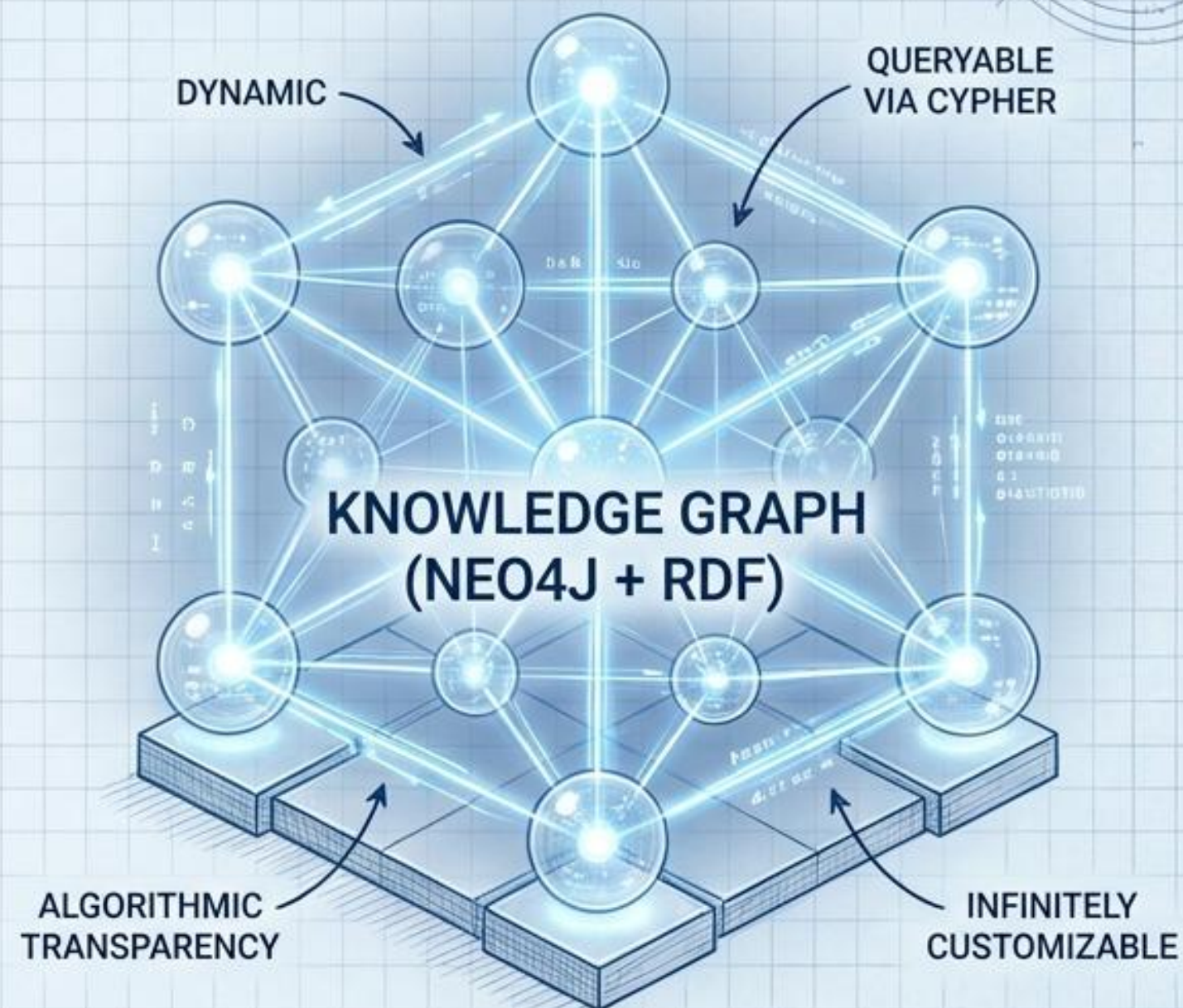
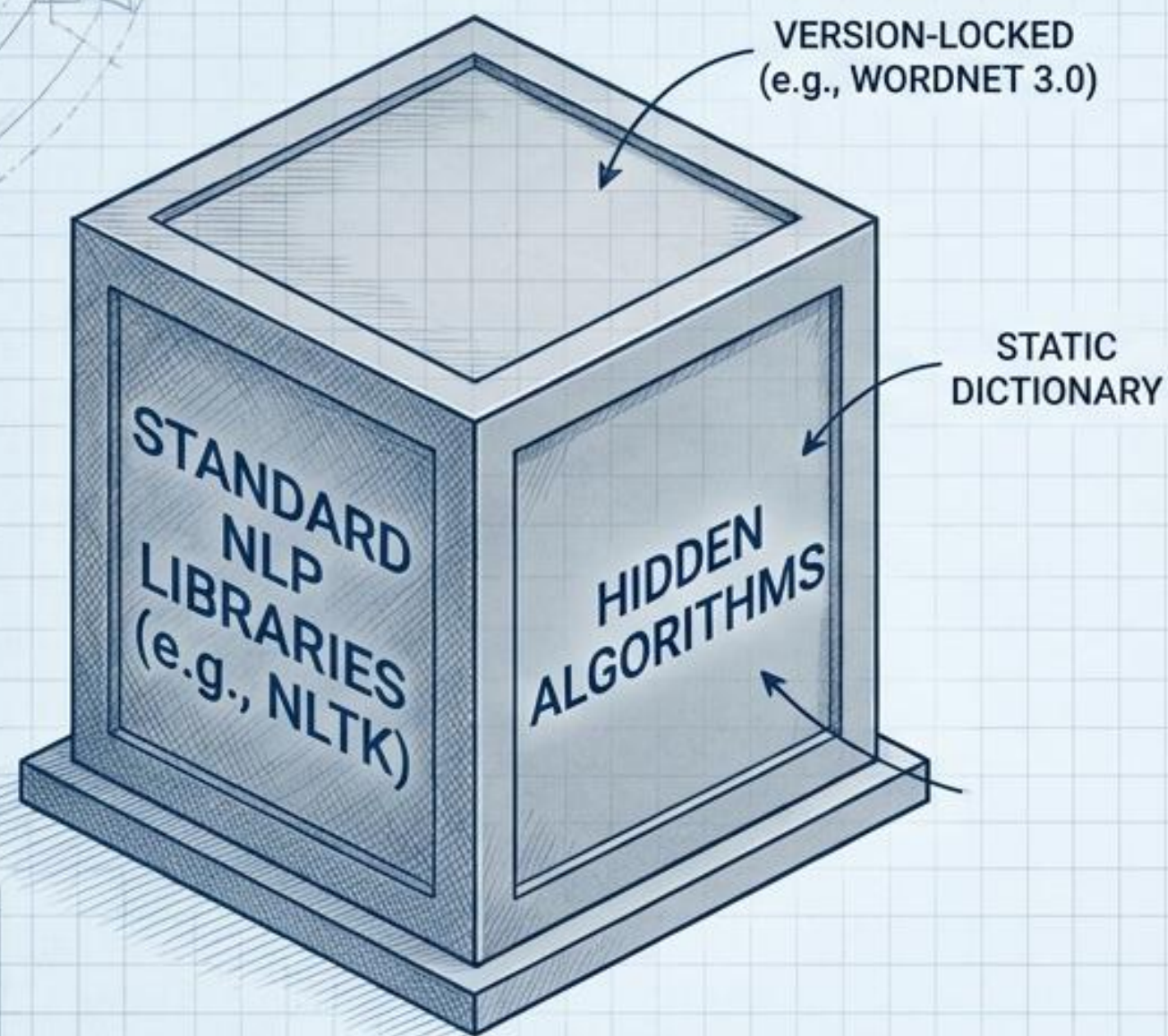
Talking to Your Knowledge Graph

A Practitioner's Guide to Lexical Databases,
Semantic Similarity, and Custom Graph Architectures



dr Zbigniew Galar

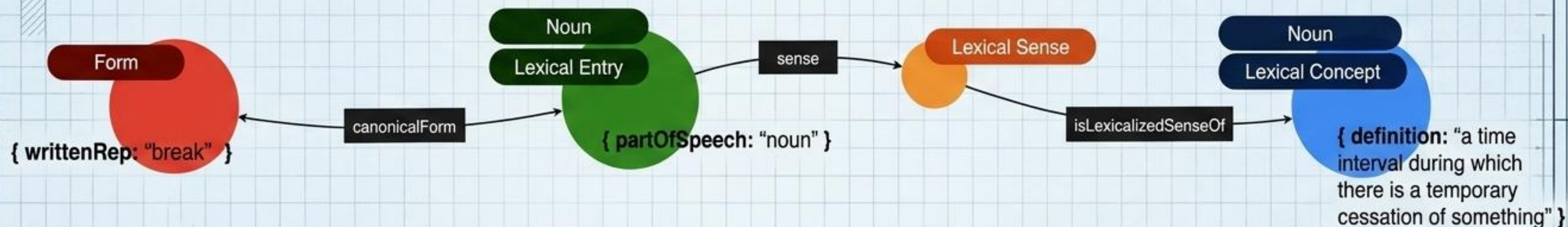
BEYOND THE BLACK BOX



WHILE STANDARD NLP LIBRARIES RELY ON STATIC, VERSION-LOCKED DEPENDENCIES, A NATIVE KNOWLEDGE GRAPH TURNS LANGUAGE INTO A MATHEMATICAL, HIGHLY CUSTOMIZABLE REALITY.

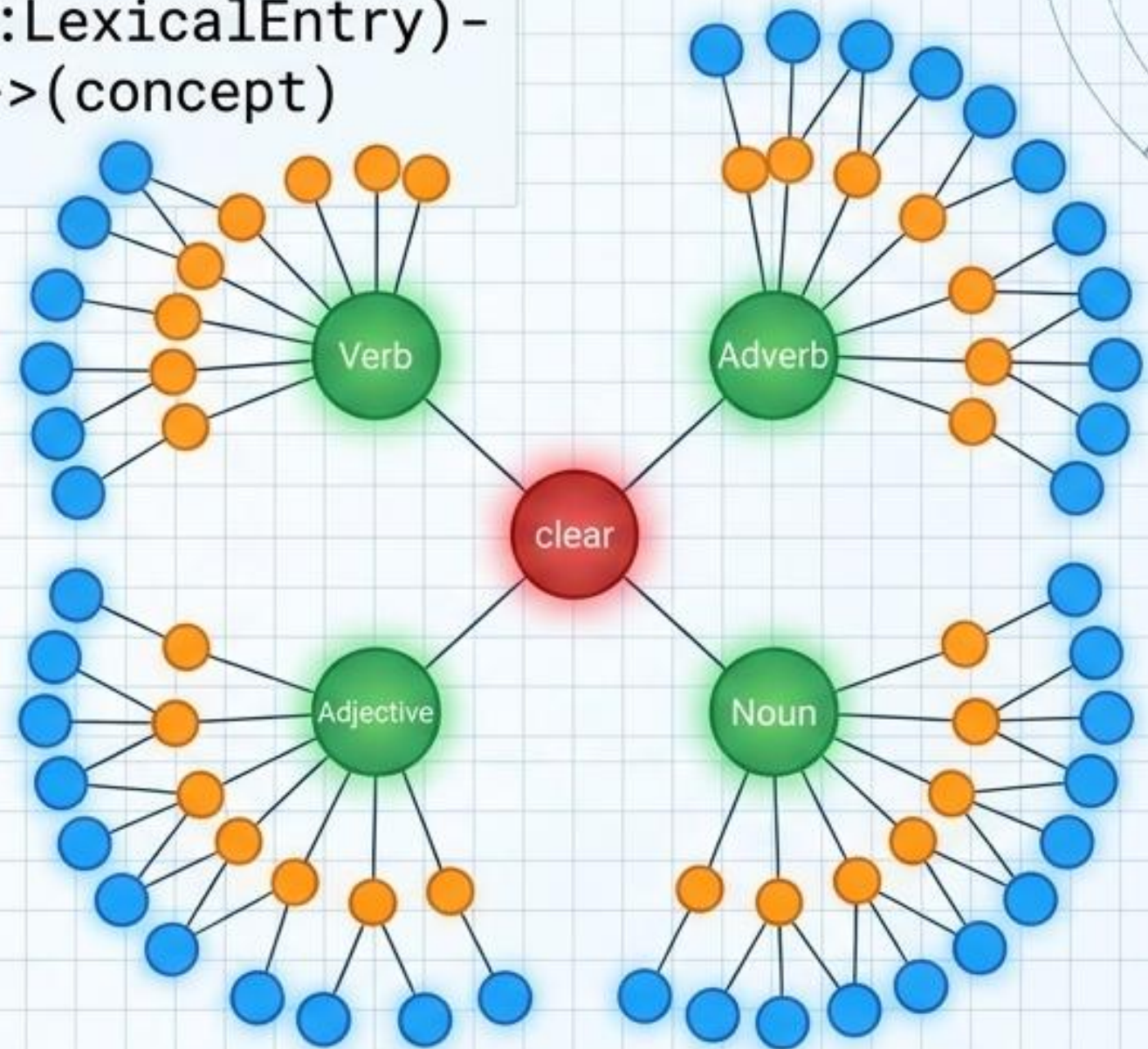
THE ANATOMY OF A LEXICAL GRAPH

WordNet is constructed on a strict **four-node taxonomy**. This schema separates the written word from its cognitive meaning, allowing us to mathematically map the nuances of language.



Querying for Meaning: One Word, Multiple Roles

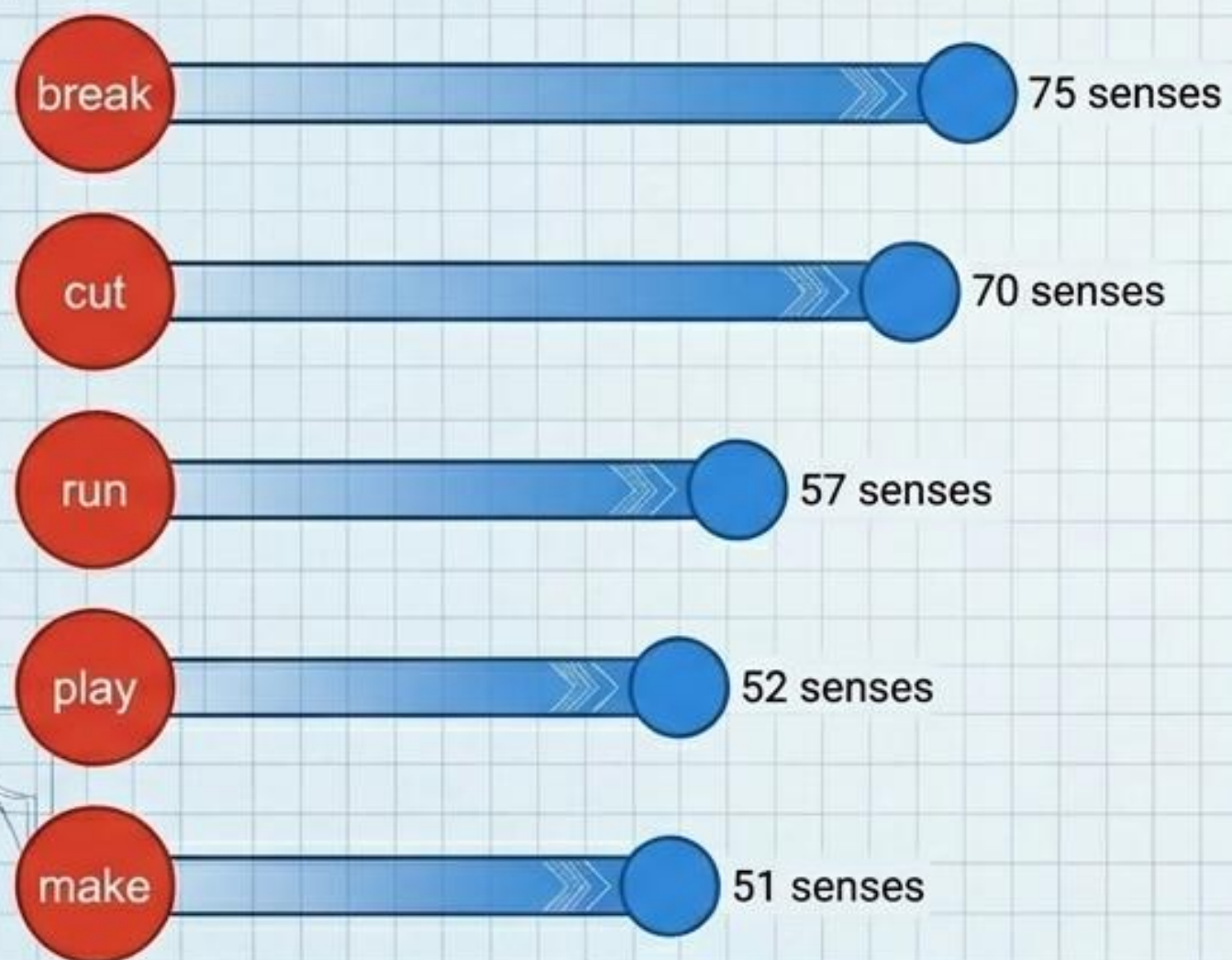
```
MATCH (lemma:Form)<-[:canonicalForm]-(le:LexicalEntry)-  
-[:sense]->()-[:isLexicalizedSenseOf]->(concept)  
WHERE lemma.writtenRep = 'clear'
```



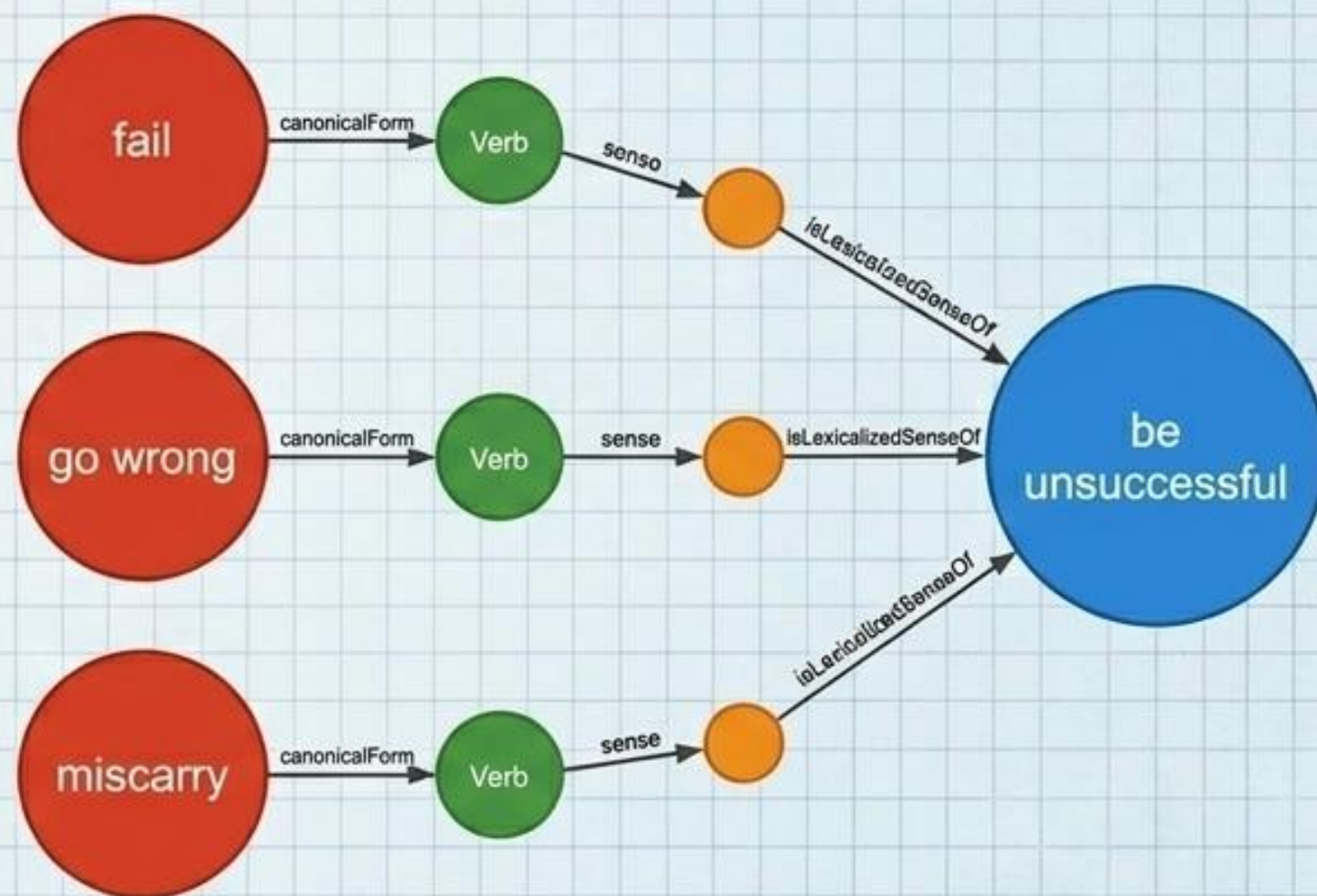
Traversal instantly reveals the vast semantic footprint of highly adaptable words.

The Extremes of Language

Polysemy (One Word → Many Meanings)

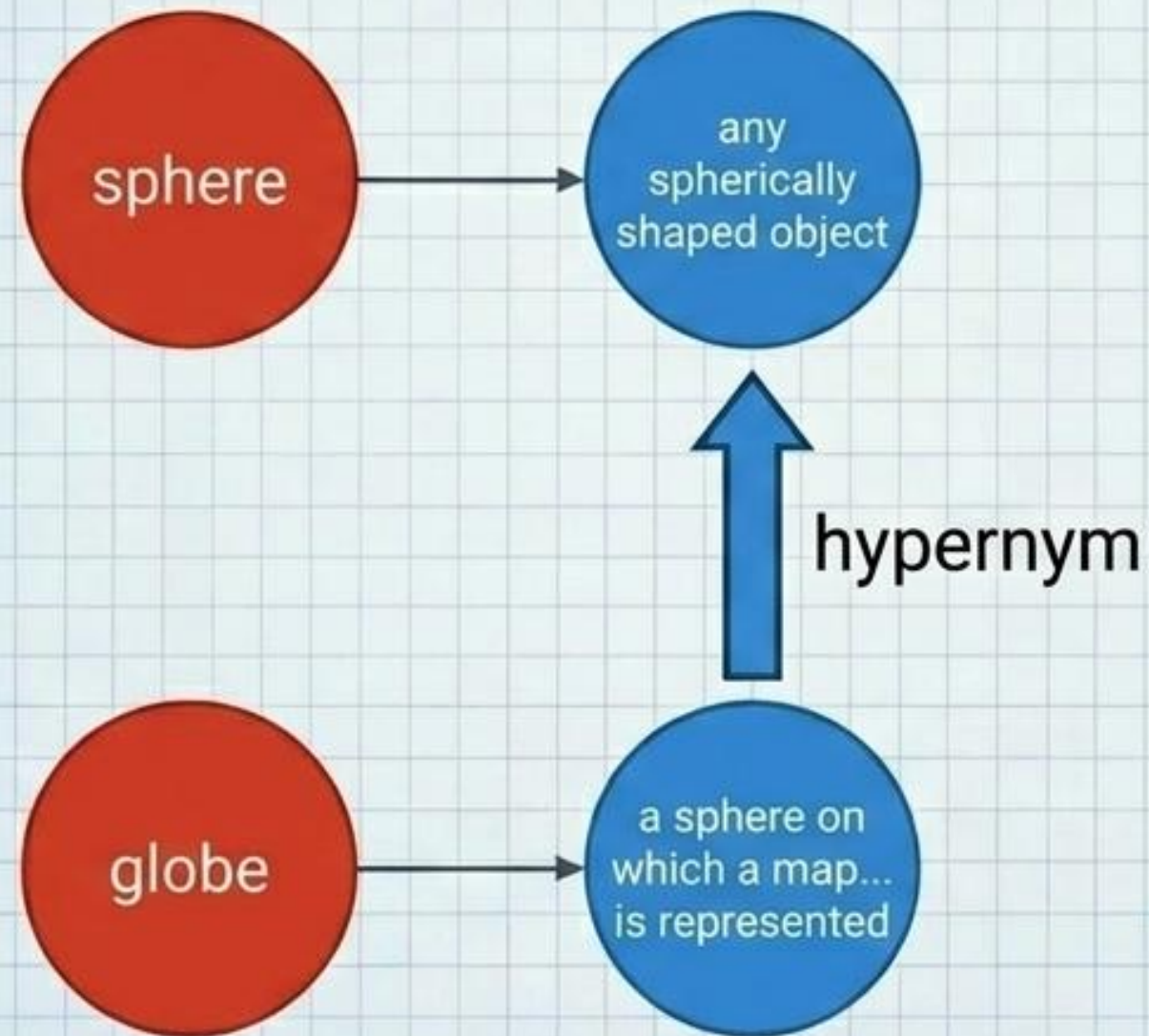


Inverse Lookup (Many Words → One Meaning)

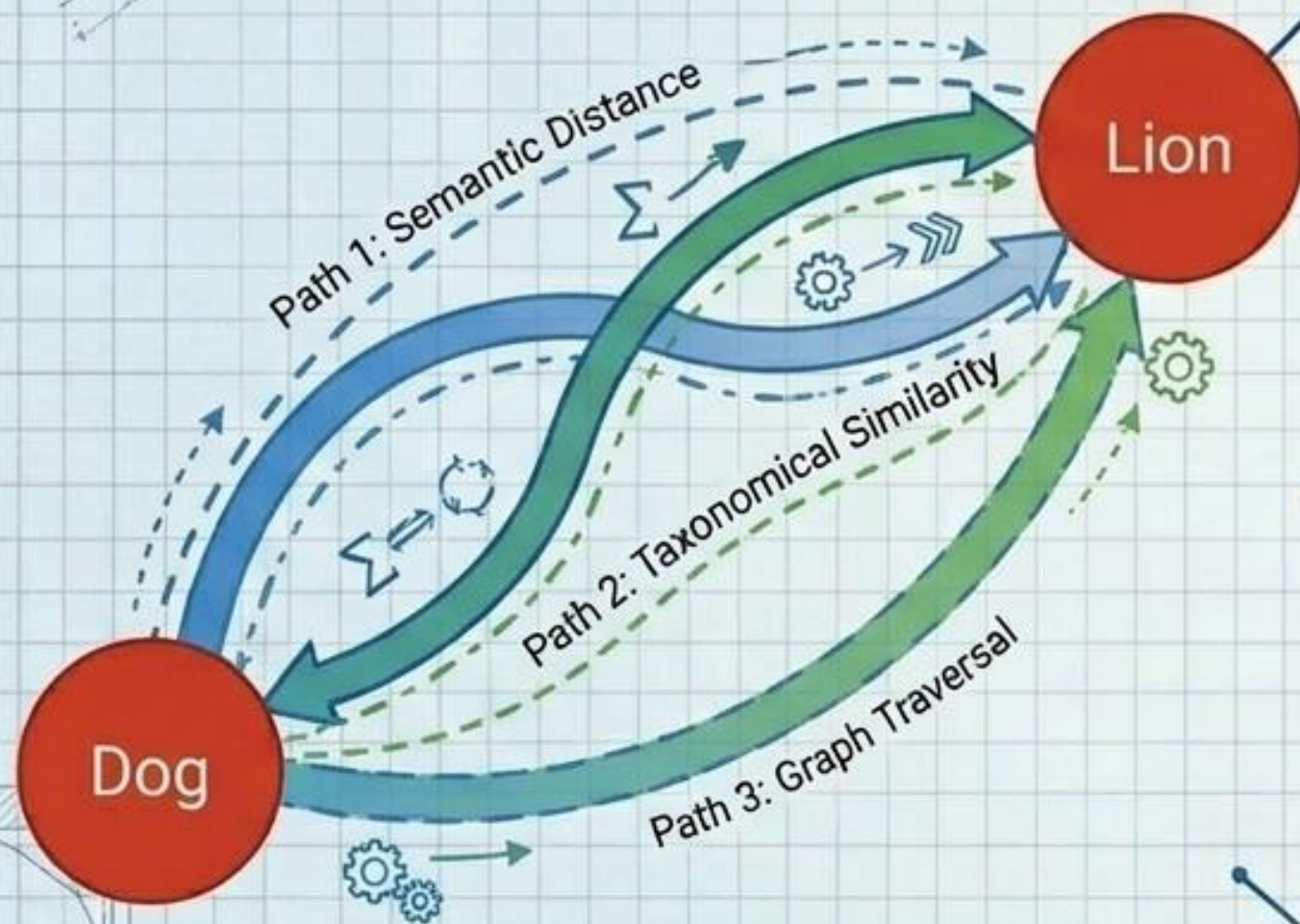


Navigating Semantic Taxonomies

The graph natively enriches concepts with semantic relationships. By traversing **hypernym** (more general) and **hyponym** (more specific) relationships, the graph maps words onto an interconnected hierarchy.



Why Measure Meaning in Cypher?



Zero Version Dependency.

Libraries like NLTK are stuck on WordNet 3.0. Graph queries run dynamically on the latest data.

Infinite Customizability.

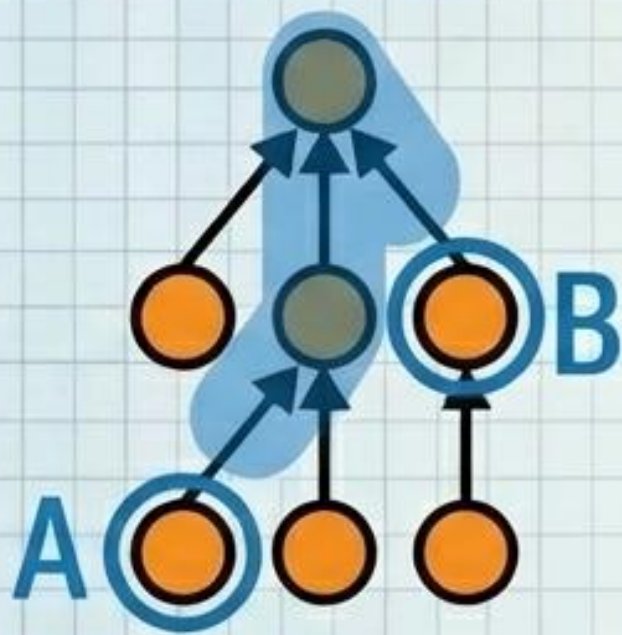
Instantly compute similarities on newly added, domain-specific vocabularies that external dictionaries don't know exist.

Algorithmic Transparency.

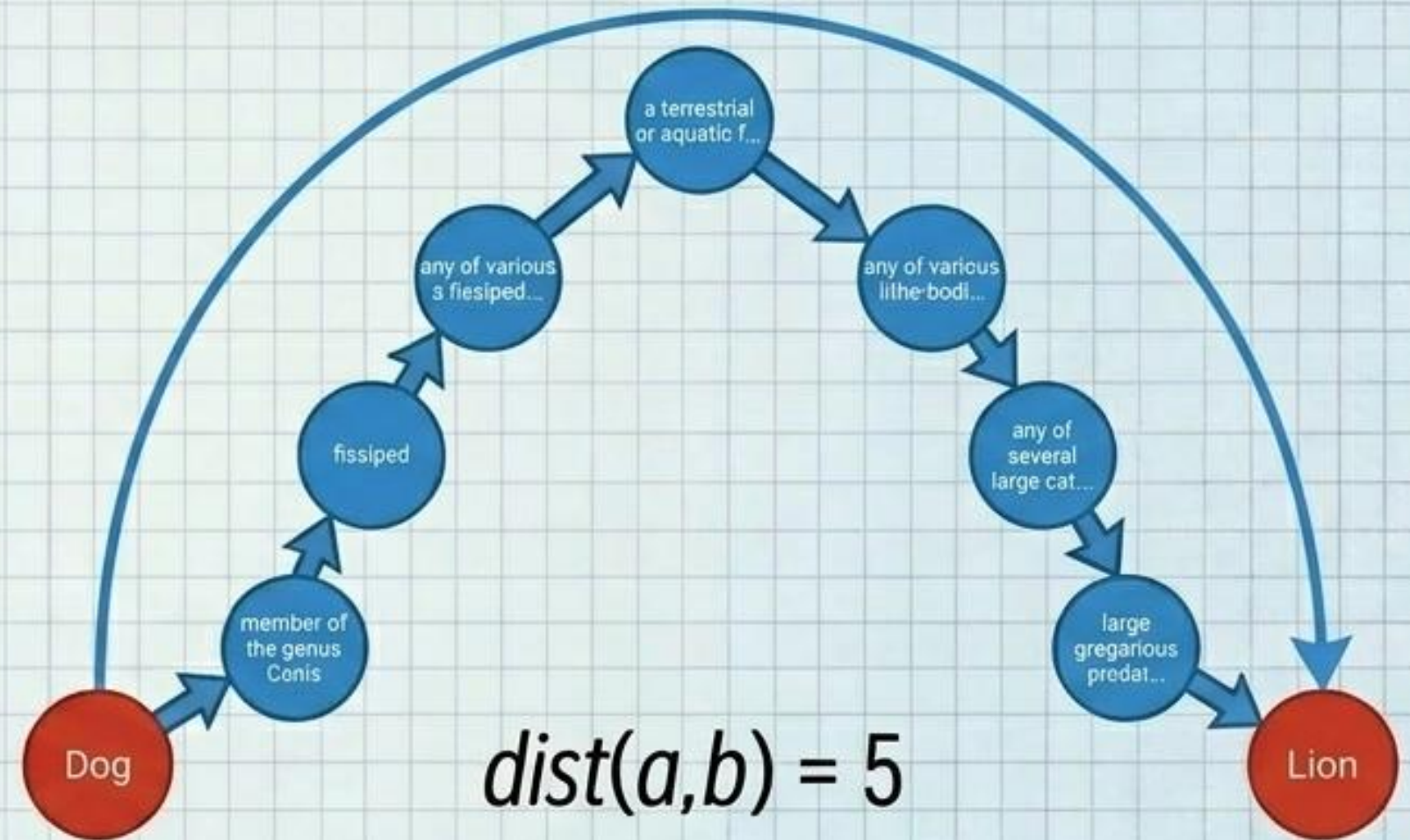
Modify, tune, and audit the mathematics of semantic similarity directly via graph traversal.

Algorithm 1: Path Similarity

Computes similarity purely based on the shortest structural path connecting two concepts in the taxonomy.



$$sim(a,b) = \frac{1}{1 + dist(a,b)}$$



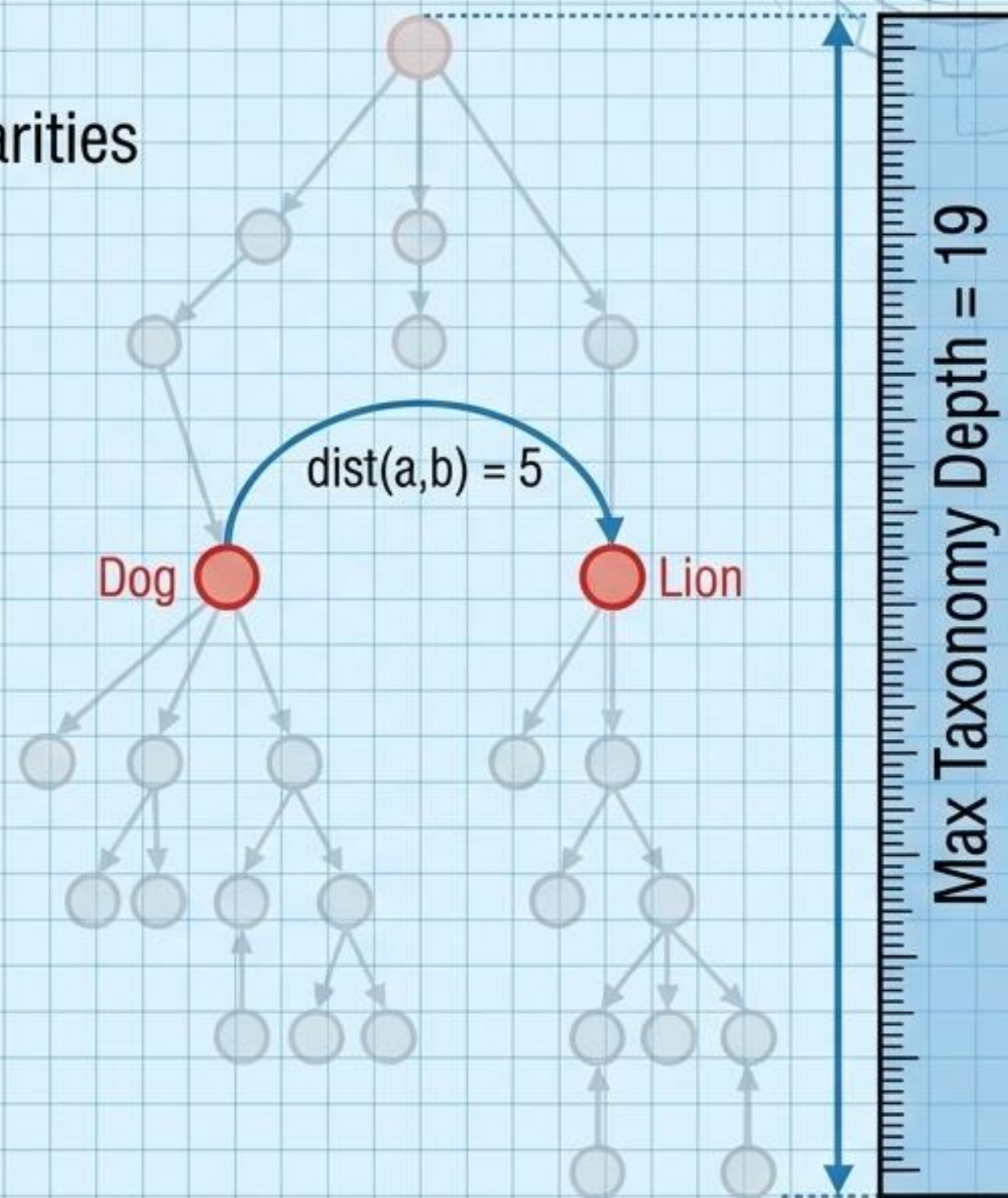
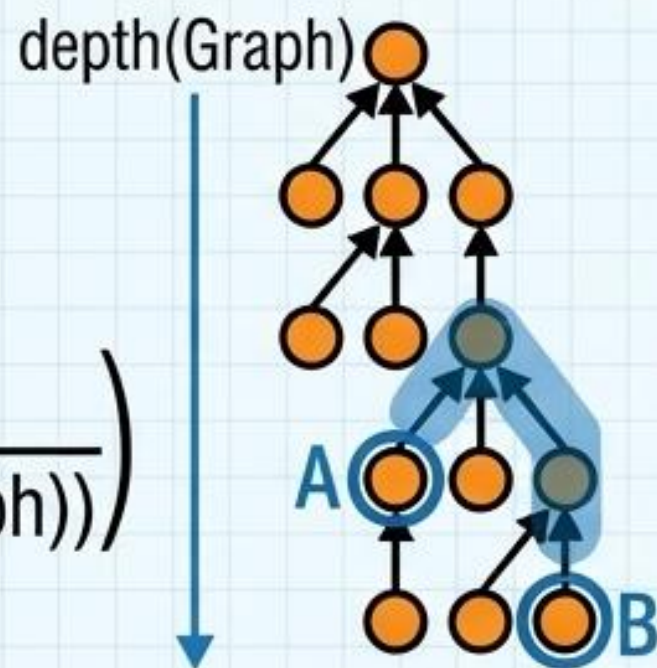
Dog vs. Lion Similarity = 0.166

Algorithm 2: Leacock-Chodorow Similarity

Introduces global context by adding the total depth of the taxonomy to the computation, significantly boosting similarities that occur in deeper, more highly-specified hierarchies.

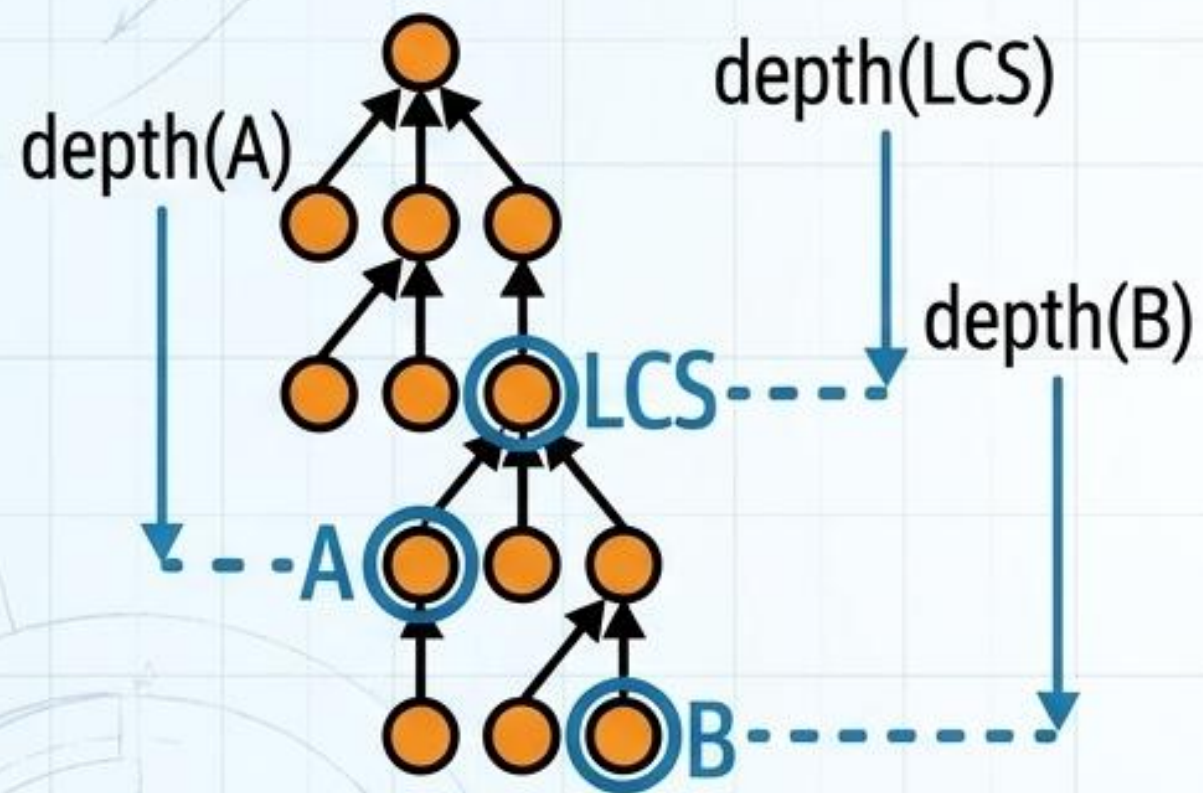
Leacock-Chodorow

$$\text{sim}(a,b) = -\log\left(\frac{\text{dist}(a,b)}{2 \times \text{depth}(\text{Graph})}\right)$$

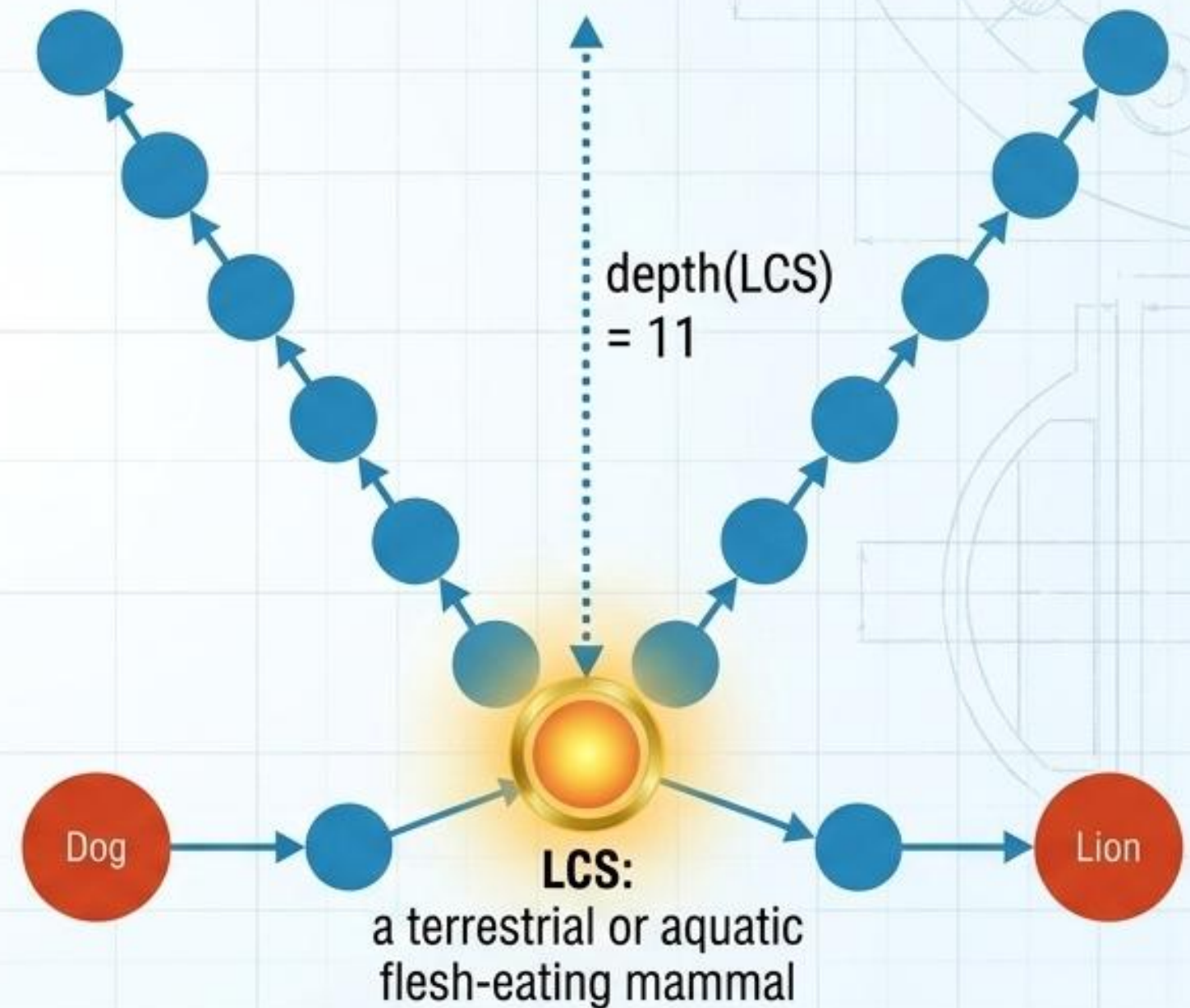


Algorithm 3: Wu-Palmer Similarity

Discards global graph depth in favor of localized specificity, focusing on the depth of the most specific ancestor node shared by the two concepts (the Least Common Subsumer).



$$sim(a,b) = \frac{2 * depth(LCS)}{depth(a) + depth(b)}$$

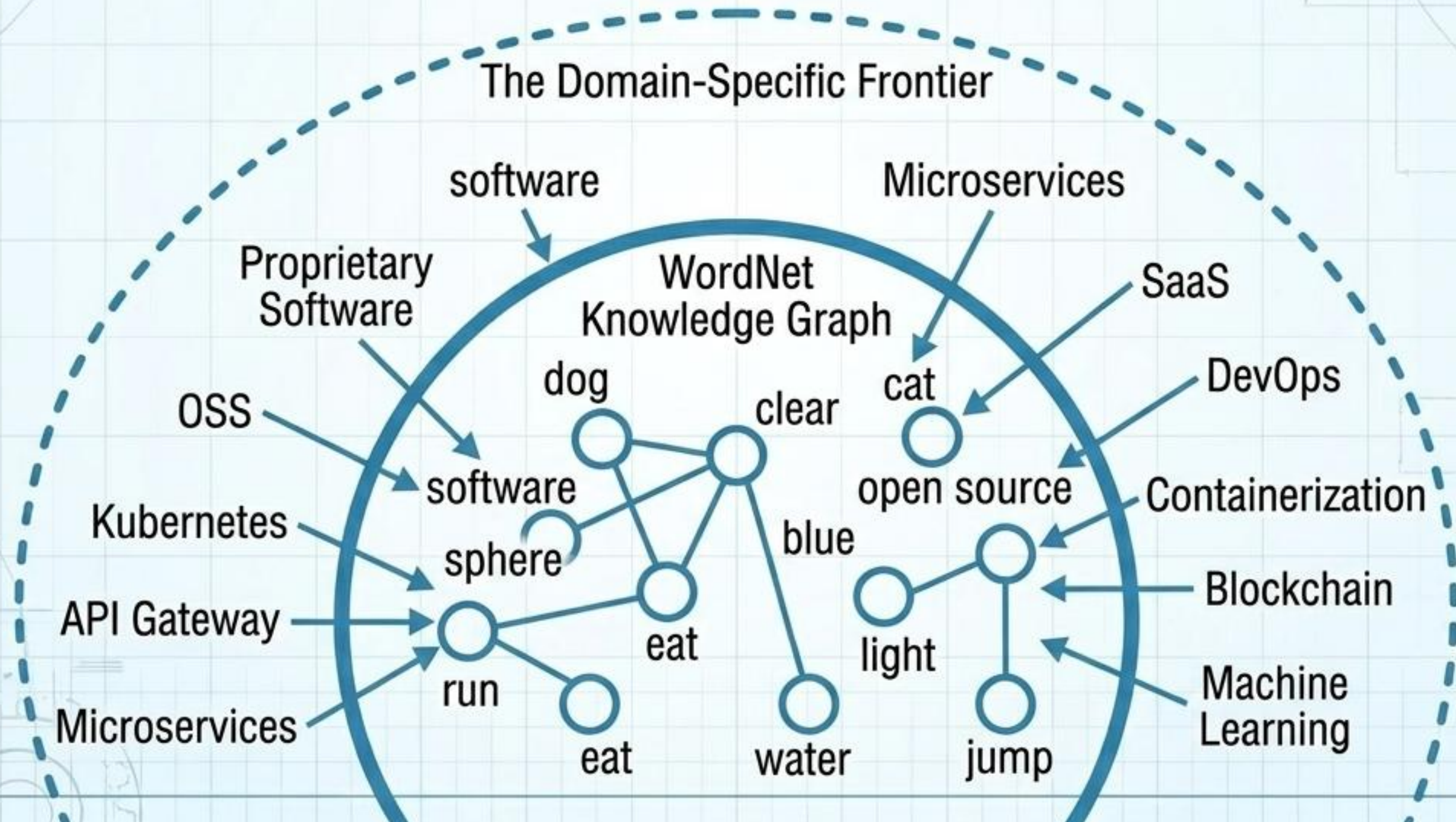


The Similarity Matrix: A Diagnostic Guide

	Path Similarity	Leacock-Chodorow	Wu-Palmer
Core Logic	Shortest distance	Distance contextualized by global graph scale	Specificity of shared ancestry
Required Graph Variables	dist	dist, Graph Depth	Depth(a), Depth(b), Depth(LCS)
Cypher Efficiency Context	Low compute, simple traversal	Requires global graph aggregation max(length)	Requires converging node matching

Escaping the Bounds of the Dictionary

Standard dictionaries map general English. However, true enterprise intelligence requires systems that understand proprietary, domain-specific terminology that external NLP libraries simply cannot recognize. The graph's dynamic architecture solves this.



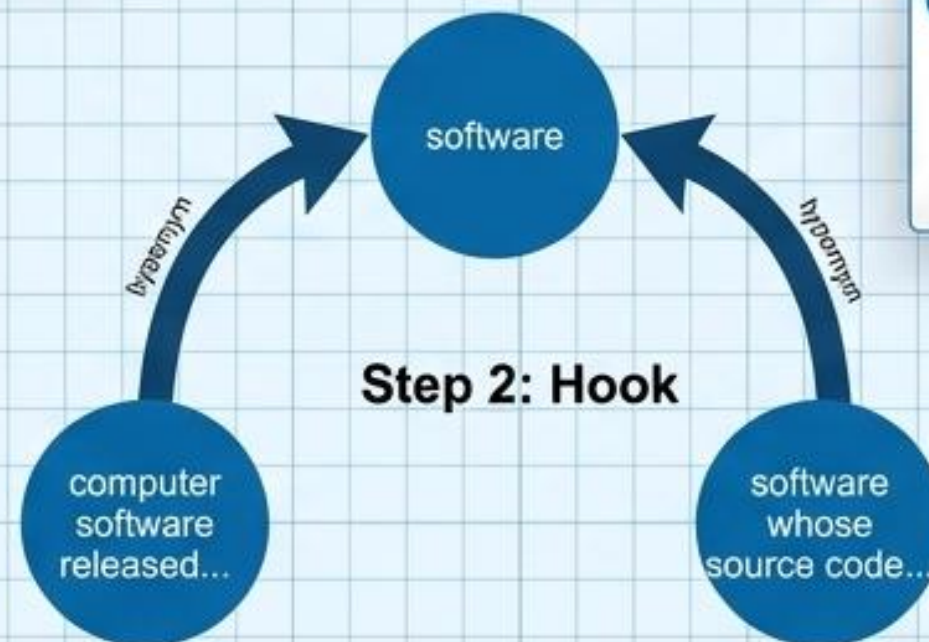
Extending the Blueprint: Injecting Domain Knowledge

By replicating the strict 4-node schema and hooking the new concepts into existing hypernym hierarchies, custom terminology instantly becomes a native part of the semantic network.

Step 1: Construct



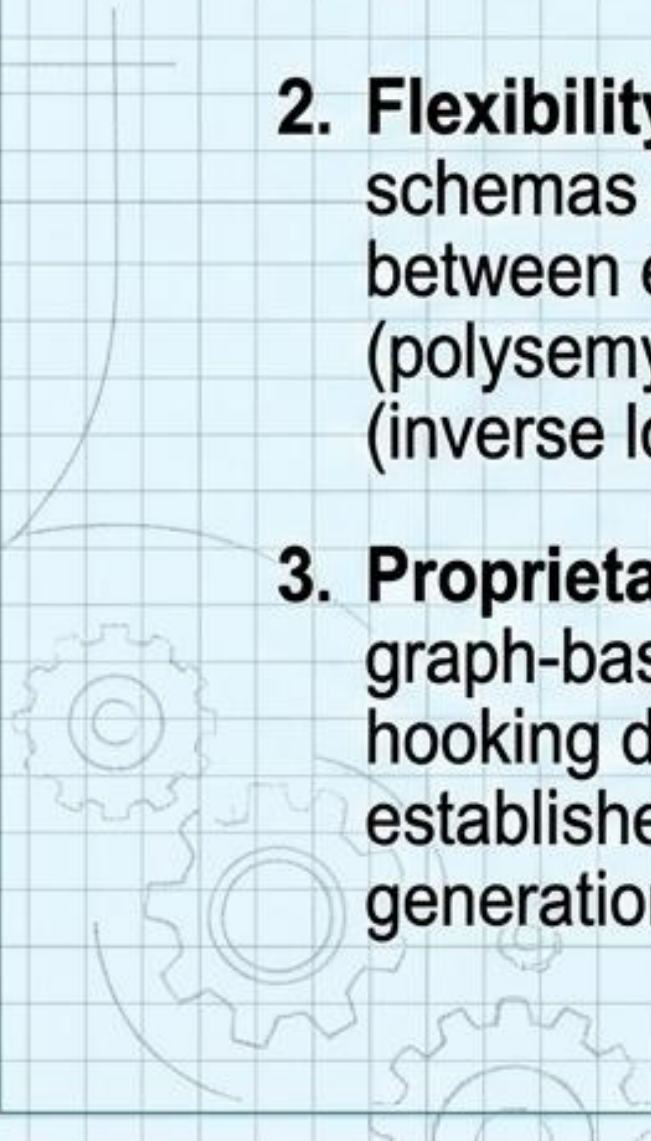
Step 2: Hook



The Aha! Synthesis: Instant Semantic Calculation

Once the new domain entities are injected, the exact same Cypher similarity algorithms run seamlessly on vocabulary that doesn't exist in standard dictionaries.

Concept A	Concept B	Path Sim	Leacock-Chodorow Sim	Wu and Palmer Sim
software whose source code is available under an open source license	computer software released under a license restricting use, study, or redistribution	0.400	0.846	0.973

- 
2. **Flexibility**
 schemas
 between e
 (polysemy)
 (inverse l
3. **Proprieta**
 graph-bas
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